Exploratory Data Analysis and PCA

From understanding data to discovering latent structure

Data-driven Systems Engineering

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EDA: ask questions before fitting models

- Imagine a dataset of patient measurements: does each row represent a patient, a visit, or a measurement?
- Before computing any statistic, ask: **what generated these observations?**
- EDA is a cycle: **question** $\rightarrow$ **summary/plot** $\rightarrow$ **possible explanation** $\rightarrow$ **new question**.
- The objective is not to find the prettiest plot, but to identify structure, data problems and potential modeling assumptions.

Key idea

A plot is evidence to investigate, not automatically a conclusion.

EDA: a roadmap for asking better questions

1. **Validate:** What does a row mean? Are values plausible and comparable?
2. **Describe:** What is typical? How variable are observations?
3. **Explore:** What shapes, groups and relationships appear?
4. **Challenge:** Could patterns reflect outliers, confounders or sampling bias?
5. **Prepare:** Which transformations are defensible before modeling?

Key idea

EDA is not a single statistical test: it is iterative investigation.

First inspect the unit of observation

- How many observations and variables?
- Are values measurements, identifiers or categorical codes?
- Are units consistent? Are some patients measured repeatedly?
- Which column, if any, is the target? It must not be treated as an ordinary input feature.

Iris example

150 flowers, four numeric measurements in cm and a species label.

A row means *one flower*, not one species.

Numeric, categorical, and ordered variables

- **Numerical:** blood pressure, age, concentration; examine range, distributions and units.
- **Nominal categorical:** treatment type, device model; examine frequencies, not means.
- **Ordinal:** disease stage or severity score; the order is meaningful, but distances may not be.
- **Identifiers:** patient ID or admission number is not a numerical predictor just because stored as an integer.

Key idea

The meaning of a variable determines which summaries and plots are appropriate.

Data quality: a plausible value may still be wrong

- Patient age $= 180$: obvious validity problem; age $= 65$: plausible but could be recorded for the wrong visit.
- Units: mixing milligrams and grams creates artificial outliers.
- Duplicates: repeat encounters may be legitimate; exact duplicate rows may not be.
- Repeated measurements create dependence: splitting rows randomly can leak patient identity.

Question

Would removing all "extreme" points discard the very clinical events we wish to detect?

Start with a few high-value checks

```
from sklearn.datasets import load_iris
iris = load_iris(as_frame=True)
df = iris.frame.copy()
print(df.shape)
print(df.dtypes)
print(df.isna().sum())
print(df.describe().T)
print(df['target'].value_counts())
```

Questions to answer

Are all four features numeric? How do their ranges differ? Are the classes equally represented? Would any missing value be surprising?

Missingness has a mechanism

- **Missing completely at random:** an occasional sensor readout is lost independently of the process.
- **Missing at random given observed data:** older instruments record a variable less often.
- **Missing not at random:** extremely high values are more likely to be unrecorded.

Why this matters

Dropping rows can change the population represented by the data. Mean imputation can reduce variability and distort correlations.

Key idea

Before fixing missing values, ask *why* they might be missing.

Mean and median answer different questions

Suppose five measurements are $[2, 3, 3, 4, 28]$.

$$\text{mean} = 8, \qquad \text{median} = 3.$$

- Mean balances all observations and is strongly affected by extreme values.
- Median is the central ordered value and is more robust to extremes.
- Standard deviation measures dispersion around the mean; IQR describes the middle 50%.

Question

Is 28 an error, a rare valid observation, or evidence of a second population? The summary statistics alone cannot tell us.

What can a histogram reveal?

- **Location:** where are the values concentrated?
- **Spread:** are values tightly packed or dispersed?
- **Skewness:** is one tail longer?
- **Multiple peaks:** could there be hidden subgroups?
- **Gaps/extremes:** sampling issue or actual structure?

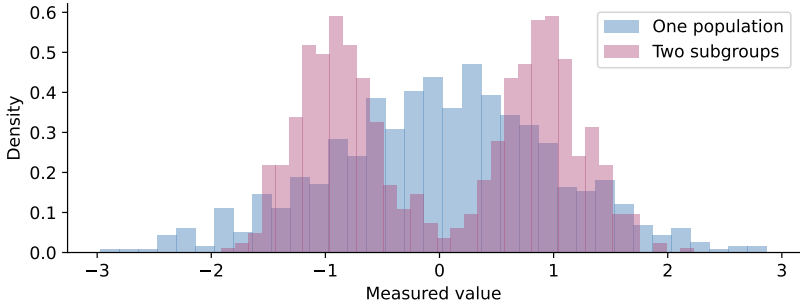
Iris thought experiment

A histogram of petal length may have more than one peak *because species are mixed*. Color by species to investigate.

Key idea

Histograms depend on bin width; a shape that appears/disappears with bins needs caution.

Same mean and variance, different data stories



Two datasets may have nearly identical mean and standard deviation while one contains two subpopulations. **Inspect the shape**, not only the first two moments.

Quantiles and robust spread: what is typical?

For sorted measurements $[2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 25]$ (median-of-halves convention):

$$Q_1 = 4, \quad \text{median} = 7, \quad Q_3 = 10, \quad \text{IQR} = 6.$$

- Middle 50% of observations fall in the box $[4, 10]$.
- Tukey fences: $Q_1 - 1.5\,\text{IQR} = -5$ and $Q_3 + 1.5\,\text{IQR} = 19$.
- Whiskers end at the *actual observed* values 2 and 12; 25 is plotted separately.

Key idea

An outlier flag is a prompt to investigate, not a rule to delete.

Boxplot: a compact summary, not an outlier detector

- Box: 25th to 75th percentiles (IQR). Horizontal line: median.
- Common whisker convention: most extreme points within $1.5\,\mathrm{IQR}$ of the quartiles.
- Points beyond whiskers are *flagged for investigation*, not automatically incorrect.

Why compare by group?

Two groups can have the same median but very different spreads. Pooled summaries may conceal this difference.

Scatterplots show what a single number hides

- Look for a direction (positive/negative), shape (linear/curved), clusters, unequal variability and influential observations.
- Plotting points by species can reveal whether a relationship holds *within* groups or only across the pooled dataset.
- A pairplot repeats this for many feature pairs; use it to select questions worth following up.

Example

A strong pooled correlation between petal length and width may partly reflect differences *between* species.

Pooled relationships can conceal group differences

- Imagine two clinics: clinic A has younger patients and low average risk; clinic B has older patients and higher average risk.
- Across clinics, age and risk can appear strongly associated even if the within-clinic association is weak or reversed.
- Inspect stratified scatterplots, conditional summaries and possible confounding variables.

Question

Are we learning a relationship between two biological variables, or primarily a difference between data sources?

Covariance: do two variables deviate together?

For paired observations, define deviations from the sample means:

$$a_i = x_i - \bar{x}, \qquad b_i = y_i - \bar{y}.$$

Then

$$s_{XY} = \frac{1}{n-1} \sum_{i=1}^{n} a_i b_i.$$

- Both above or both below their means: $a_i b_i > 0$.
- One above and the other below: $a_i b_i < 0$.
- Averaging products measures their tendency to co-vary.

Key idea

Covariance has units (X units)$\times$(Y units), so its magnitude is scale-dependent.

Pearson correlation: normalized covariance

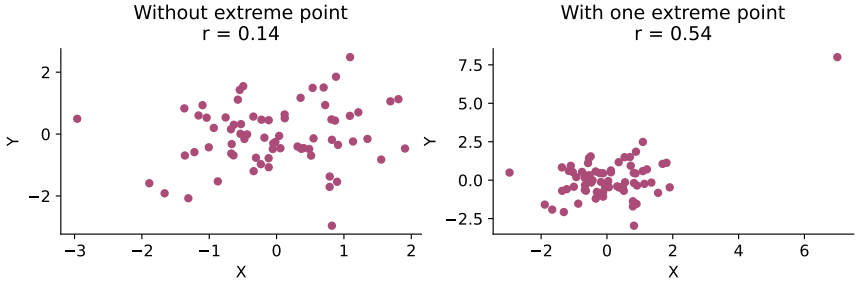
$$r_{XY} = \frac{s_{XY}}{s_X s_Y} = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2}\sqrt{\sum_i (y_i - \bar{y})^2}}$$

- $r \approx +1$: centered values move in the same linear direction.
- $r \approx -1$: one grows as the other falls along a line.
- $r \approx 0$: little *linear* association; nonlinear dependence may still exist.

Geometric interpretation

r is the cosine of the angle between the two *centered observation vectors* in $\mathbb{R}^n$.

One influential point can change Pearson correlation

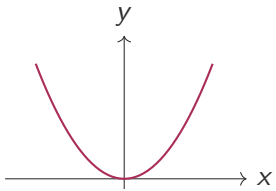


Pearson correlation is sensitive to influential observations. Before claiming a strong association, inspect the scatterplot and consider group structure.

Correlation zero does not mean independence

Let X be symmetric around zero and $Y = X^2$.

- Y is completely determined by X.
- Yet
  $\mathrm{Cov}(X, Y) = E[X^3] - E[X]E[X^2] = 0$
  (under symmetry).
- Pearson correlation is zero, although dependence is strong and nonlinear.



Key idea

Never interpret a correlation matrix without checking scatterplots.

A heatmap is a starting point, not an explanation

- A correlation heatmap helps locate groups of approximately redundant numerical features.
- Each cell reports *pairwise linear* association: interactions, nonlinearities and causal mechanisms are invisible.
- Strong correlations may arise from shared units, derived measurements or mixing populations.
- Check individual scatterplots before selecting features to remove.

Key idea

Pattern discovery should lead to an explanation to test, not a causal claim.

EDA with a prediction target: ask different questions

- Classification: class balance, distributions by label, overlap between classes.
- Regression: target distribution, heteroscedasticity, outliers and nonlinear trends.
- Check whether apparent predictors are measured *after* the outcome (target leakage).
- Avoid using the held-out test set to tune data-dependent preprocessing decisions.

Example

A feature with low overall variance can still separate the classes almost perfectly. This will matter for PCA.

From correlation to redundancy

- Suppose two instruments measure approximately the same underlying quantity.
- Their scatterplot forms a long narrow cloud: most variation follows *one direction*.
- Instead of keeping two raw coordinates, can we describe position along that dominant direction with just one number?

The bridge to PCA

EDA *detects* correlated features. PCA *constructs* new axes adapted to their joint variation.

EDA checklist: before dimensionality reduction

1. Interpret data types, units, row meaning, sampling and missingness.
2. Examine shapes, quartiles, outliers and subgroup differences.
3. Inspect bivariate plots and correlation structure.
4. Decide whether redundancies are noise, meaningful measurements or both.
5. State what we want from reduction: visualization, compression, denoising or prediction.

What is the goal of PCA dimensionality reduction?

Given n observations in d dimensions, find a representation in $k < d$ dimensions:

$$\underbrace{X \in \mathbb{R}^{n \times d}}_{\text{original measurements}} \longrightarrow \underbrace{Z \in \mathbb{R}^{n \times k}}_{\text{compact coordinates}} .$$

- Create k new *linear combinations* of the original features.
- Preserve as much sample variance as possible under orthogonal projection.
- Equivalently, minimize the sum of squared reconstruction distances.
- The new coordinates can help visualize, compress, or reduce redundancy.

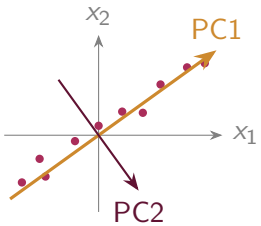
Key idea

PCA preserves variance, not labels, clinical meaning or every kind of structure.

PCA: find the best direction for a shadow

Imagine projecting a cloud of 2D points onto a line.

- A poorly chosen line makes the projected points overlap.
- A line aligned with the cloud preserves their separation.
- **PC1** is the line onto which projected data have maximum variance.
- **PC2** is perpendicular and preserves remaining variability.



PCA changes coordinates; reduction happens when components are discarded.

How do we compute PCA? The recipe

1. Subtract each feature mean: $X_c = X - \mathbf{1}\mu^\top$.
2. Calculate the covariance matrix: $S = \frac{1}{n-1} X_c^\top X_c$.
3. Find eigenvalues and eigenvectors: $Sw_j = \lambda_j w_j$.
4. Sort directions by decreasing eigenvalue and choose the first k.
5. Project the observations: $Z = X_c W_k$.

Compute the covariance matrix

For the centered data X_c:

$$\boxed{S = \frac{1}{n-1} X_c^\top X_c \in \mathbb{R}^{d \times d}}$$

For two features:

$$S = \begin{pmatrix} \mathrm{Var}(X_1) & \mathrm{Cov}(X_1, X_2) \\ \mathrm{Cov}(X_1, X_2) & \mathrm{Var}(X_2) \end{pmatrix}.$$

- Diagonal = how much each original feature varies.
- Off-diagonal = how much pairs of features move together.

Key idea

The covariance matrix summarizes the orientation and spread of a linear data cloud.

Find a direction of maximum projected variance

Choose a unit vector $w \in \mathbb{R}^d$. Project all observations:

$$z = X_c w \in \mathbb{R}^n.$$

We can calculate how spread out these projected coordinates are:

$$\mathrm{Var}(z) = \frac{z^\top z}{n-1} = \frac{w^\top X_c^\top X_c w}{n-1} = \boxed{w^\top S w}.$$

Optimization question

Which unit direction w maximizes $w^\top S w$? The unit-length constraint avoids increasing variance simply by multiplying w by a large number.

Solve for the principal directions

We maximize $w^\top S w$ subject to $w^\top w = 1$. Introduce a multiplier λ:

$$\mathcal{L}(w, \lambda) = w^\top S w - \lambda(w^\top w - 1).$$

Setting the derivative with respect to w to zero gives

$$2Sw - 2\lambda w = 0 \qquad \Longrightarrow \qquad \boxed{Sw = \lambda w}.$$

- w = a direction that the covariance transformation does not rotate.
- $\lambda = w^\top S w$ = variance captured in that direction.
- Largest eigenvalue $\rightarrow$ first principal component.

Rank the eigenvectors and project

Let $\lambda_1 \geq \cdots \geq \lambda_d \geq 0$ with corresponding orthonormal eigenvectors $w_1, \ldots, w_d$.

$$W_k = [w_1 \ \cdots \ w_k] \in \mathbb{R}^{d \times k}, \qquad \boxed{Z = X_c W_k \in \mathbb{R}^{n \times k}}.$$

- Each row of Z is an observation in the new coordinate system.
- Each column of Z is one *principal-component score* across all observations.
- Choosing $k < d$ means keeping the dominant directions and discarding the others.

Key idea

From d measurements to k new coordinates per observation.

What do explained variance ratios mean?

$$r_j = \frac{\lambda_j}{\sum_{\ell=1}^{d} \lambda_\ell}, \qquad R_k = \sum_{j=1}^{k} r_j.$$

Toy example

Four component variances: $[7, 2, 0.7, 0.3]$.

PC1 retains 70% of the total variance. PC1+PC2 retain 90%.

- Scree plot: individual contributions r_j.
- Cumulative plot: amount retained for each k.
- 90% retained variance is *not* 90% predictive accuracy or 90% preserved information in every sense.

What is lost? Reconstruction makes it concrete

From the retained scores, reconstruct the data approximately:

$$\widehat{X} = ZW_k^\top + \mathbf{1}\mu^\top.$$

- Projection keeps the position along the first k axes.
- Reconstruction maps back to the original coordinates, but without discarded directions.
- Error can be measured by $\|X - \widehat{X}\|_F^2$.

Important equivalence

Among k-dimensional linear subspaces, PCA both *maximizes retained variance* and *minimizes total squared reconstruction error*.

Scores vs. component coefficients

- **Component direction:** w_1 is a vector of coefficients, one weight for each original feature.
- **Score:** $z_{i1} = (x_i - \mu)^\top w_1$ tells where observation i lies on PC1.
- Large absolute coefficients suggest which scaled measurements contribute most to that axis.

Interpret carefully

A component may mix petal width, petal length and sepal measures. It is *not* an original physical measurement. Changing w to $-w$ flips scores but describes the same axis.

From the mathematics to Python

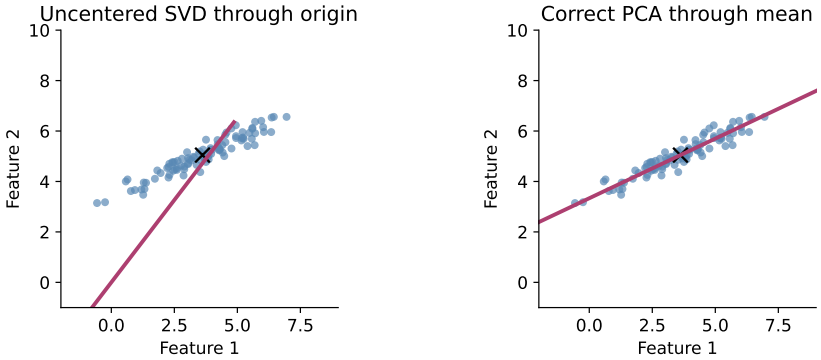
```
from sklearn.decomposition import PCA

X = iris.data                  # n x d
pca = PCA(n_components=2)
Z = pca.fit_transform(X)     # n x 2
print(pca.components_)         # 2 x d: directions
print(pca.explained_variance_ratio_)
```

Connect code to math

`PCA.fit` automatically centers the features; `components_` stores directions as rows, and `Z` contains the scores.

Example 1: what if we forget to center?



Ordinary PCA centers the data. Taking the leading singular vector of the *uncentered* matrix instead finds an axis through the origin, dominated partly by the mean offset.

Why centering changes the direction

Write each observation as $x_i = \mu + (x_i - \mu)$. Then

$$\sum_i x_i x_i^\top = \underbrace{\sum_i (x_i - \mu)(x_i - \mu)^\top}_{\text{scatter around mean}} + \underbrace{n\mu\mu^\top}_{\text{mean offset}} .$$

- Without centering, the leading singular direction may mostly point toward the mean.
- PCA on centered data instead captures directions of *variation around the mean*.
- SVD implementations can still implement PCA correctly if they center internally (as scikit-learn does).

Why must we center before PCA?

For n rows and d features, $X \in \mathbb{R}^{n \times d}$:

$$\mu_j = \frac{1}{n} \sum_i X_{ij}, \qquad X_c = X - \mathbf{1}\mu^\top.$$

- Each column of X_c has mean zero.
- Principal directions now describe *variation around the center*, not the distance from the origin.
- For any unit direction w, a projected coordinate is $z_i = (x_i - \mu)^\top w$.

Intuition

First move the center of the cloud to the origin. Then ask in which direction the cloud is most stretched.

Scaling determines what “large variance” means

- A feature recorded in euros can have much greater numerical variance than one recorded in centimeters.
- With raw covariance PCA, the larger numerical scale may dominate the first component.
- Standardization $X'_{ij} = (X_{ij} - \mu_j)/s_j$ gives each feature unit variance.

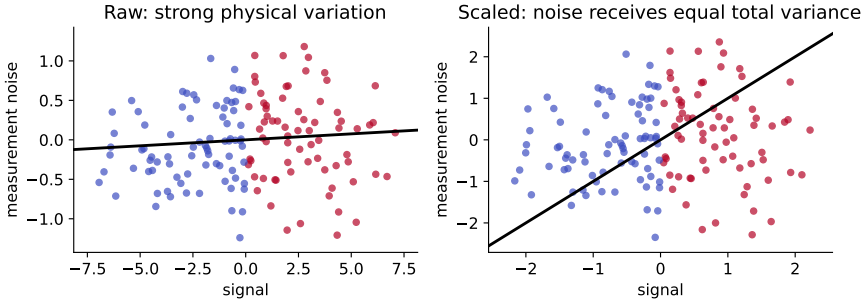
Interpretation

PCA on centered raw features uses *covariances*. PCA on standardized features is equivalent to PCA on their *correlation matrix* (apart from the convention used for SD).

Key idea

Scale before PCA when differences in units should not decide importance.

Example 2: when standardization can be harmful



Here feature 1 contains a meaningful high-variance physical signal, while feature 2 is measurement noise. Unit-variance scaling can give uninformative noise undue influence.

Covariance PCA or correlation PCA?

$$S = \frac{1}{n-1} X_c^\top X_c, \qquad Z = X_c D^{-1}, \qquad \mathrm{Cov}(Z) = D^{-1} S D^{-1}.$$

- Raw centered PCA retains the *physical relative scale* of the features.
- Standardized PCA assigns unit variance to each nonconstant feature and emphasizes relative co-variation.
- Scaling is appropriate when units are arbitrary or genuinely incomparable.
- It may be inappropriate when absolute variability has domain meaning, or very low-variance features are mostly noise.

Key idea

Scaling is a modeling assumption, not an automatic improvement.

Example 4: PCA may worsen predictive performance

Synthetic classification: class information lives in a low-variance direction. Keeping only PC1 retains a lot of variance yet discards the discriminative signal.

Why explained variance is not predictive utility

- PCA solves $\max_{\|w\|=1} \mathrm{Var}(X_c w)$ *without using labels*.
- A high-variance direction may describe noise, individual variation or nuisance factors.
- A small-variance direction may encode an accurate decision boundary.
- Compare end-to-end validation performance, not only scree plots or thresholds such as 95% explained variance.

Safe evaluation

Split data first, fit scaler/PCA on training folds only, then evaluate the whole pipeline on held-out data.

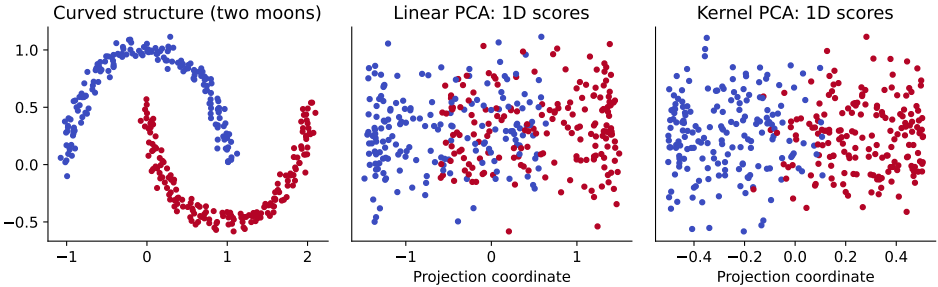
PCA is not optimized for prediction

- PCA does not use class labels or target values.
- A direction with huge variance may mostly capture irrelevant measurement variation.
- A low-variance direction can still separate two classes extremely well.
- A 2D PCA plot is useful for exploring classes, but separation is not guaranteed.

Key idea

Maximizing variance $\neq$ maximizing discrimination. Judge PCA by the task it should support.

Example 3: curved structure defeats linear PCA



On the two-moons dataset, no single straight line follows both curved groups. An RBF kernel PCA can find a nonlinear representation with better separation, depending on its kernel parameters.

What kernel PCA changes

- Linear PCA searches for axes w in the original feature space: $z = X_c w$.
- Kernel PCA implicitly maps observations through a nonlinear feature map $\phi(x)$ and computes PCA in that feature space.
- Its Gram matrix has entries $K_{ij} = k(x_i, x_j)$; for the RBF kernel,

$$k(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right).$$

- Kernel choice and γ matter; nonlinear structure is not automatically meaningful.

Key idea

A nonlinear transformation can reveal geometry that no fixed linear subspace captures.

PCA is linear — when might it fail?

- PCA finds a *flat* subspace spanned by straight orthogonal axes.
- A curved low-dimensional structure can require multiple linear components to reconstruct well.
- Nonlinear methods (kernel PCA, autoencoders, UMAP) capture different notions of structure; UMAP is mainly used for visualization.

Discussion

If data lie on a circle in two dimensions, can one straight axis preserve the circle without distortion? No.

Don't leak information during preprocessing

Training data $\rightarrow$ fit Scaler $\rightarrow$ fit PCA $\rightarrow$ fit Classifier

New data $\rightarrow$ same Scaler $\rightarrow$ same PCA $\rightarrow$ predict

- Standardization uses estimated means/variances; PCA uses estimated directions.
- Estimating either with test data leaks information across the split.
- Use a pipeline, and compare against a model that uses original scaled features.

A leakage-safe pipeline

```
from sklearn.pipeline import make_pipeline
from sklearn.linear_model import LogisticRegression

model = make_pipeline(
    StandardScaler(),
    PCA(n_components=2),
    LogisticRegression(max_iter=1000)
)
model.fit(X_train, y_train)
y_pred = model.predict(X_test)
```

Exercises: from EDA to PCA

1. With Iris: explain which plots show separation between species and which features are redundant.
2. With Wine: compare a covariance heatmap and a correlation heatmap. Why do they tell different numerical stories?
3. Implement a scaler and PCA from scratch: compute means, standard deviations, covariance, eigenpairs, scores.
4. Compare PCA with/without standardization; find k for 90% variance and evaluate reconstruction.
5. Compare a classifier trained on all features versus the first k PCs.

Decide before applying PCA

Question	Consequence
Is the mean far from zero?	Center before finding principal directions.
Are units comparable?	Consider scaling; justify its meaning.
Is the manifold curved?	Compare nonlinear alternatives or preserve original features.
Is the task predictive?	Validate task performance with and without PCA.
Is interpretation essential?	Original features may be preferable to latent components.

Key idea

PCA is a useful representation choice, not a universal preprocessing step.

Summary: a sequence of questions

1. **EDA:** What are the observations? Are distributions, groups and relationships plausible?
2. **Redundancy:** Do multiple features vary together?
3. **PCA:** Which orthogonal directions retain the largest variance?
4. **Compression:** How much variance and reconstruction accuracy do we retain with k components?
5. **Validation:** Does the representation help the task on unseen data?